GOAL-BASED AI AGENT | RAG ARCHITECTURE | FASTAPI + GEMINI + NUMPY | BAUST CSE 3202

সেবা সহায়ক (Seba Sohayok) — System Architecture

Engineering & Architectural Specification for Automated Citizen Service Assistant

Course: CSE 3202 — Artificial Intelligence Sessional | Scope: Passport & Driving Licence Services (Bangladesh)

1. Executive Summary & Design Principles

Seba Sohayok is a goal-based conversational agent engineered to navigate citizens through administrative government procedures in Bangladesh (Passport and Driving Licence services). Unlike traditional chatbots that generate speculative responses, this system is governed by three architectural invariant rules:

🛡️ 1. Zero Hallucination

The LLM (Gemini) is strictly forbidden from generating fees, deadlines, documents, or office names. All factual cards are rendered verbatim from verified JSON knowledge entries.

⚡ 2. Bimodal Speed Paths

Action chip clicks bypass the LLM completely, fetching answers directly from memory in ~5ms. Free-text uses an intent-and-slot extractor with conversation clarification.

🎯 3. Confidence Refusal

If semantic retrieval scores below the confidence threshold (`CONF_THRESHOLD = 0.62`), the agent explicitly refuses rather than generating an unreliable guess.

2. End-to-End System Architecture

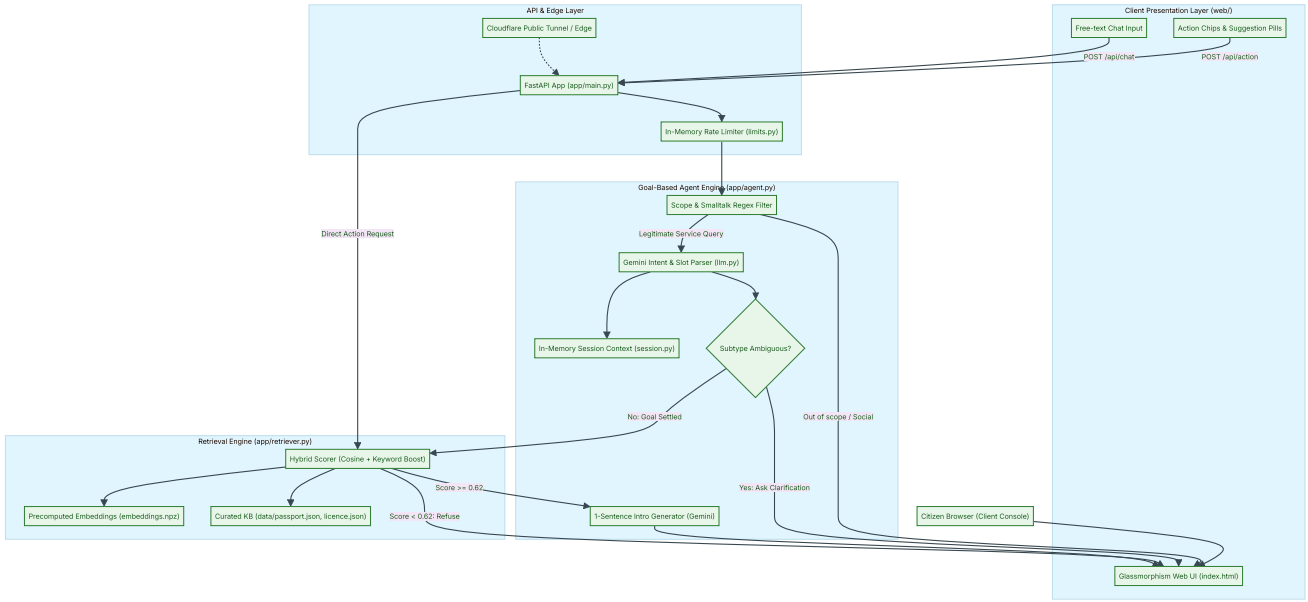


Figure 1: High-level architectural data flow showing dual-path execution (Fast Action vs. Cognitive Chat).

3. Execution Sequence Diagrams

3.1 Fast-Path Execution (Action / Chip Click — Sub-10ms)

When a citizen clicks an action chip, the intent and slot are already unambiguous. The LLM is bypassed completely:

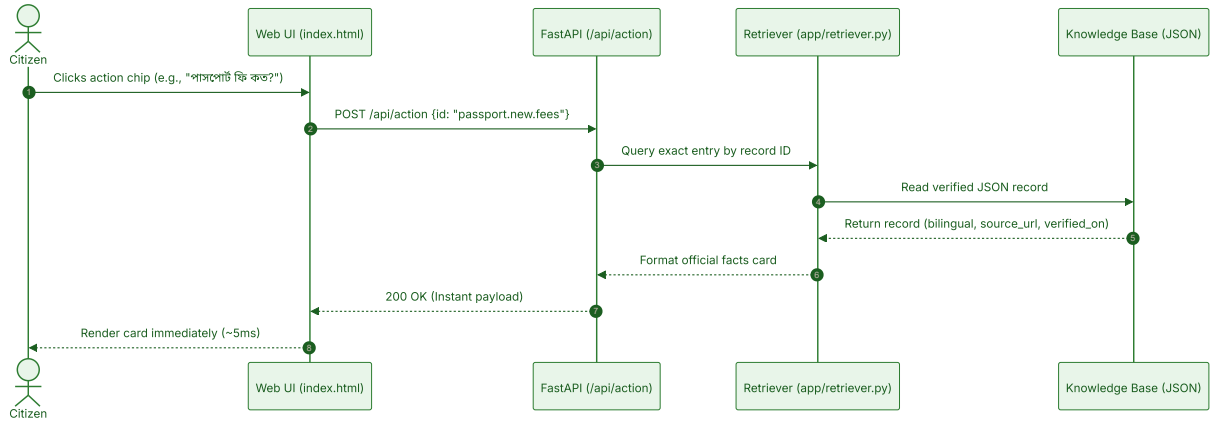


Figure 2: Deterministic, LLM-free fast path for direct action clicks.

3.2 Cognitive Free-Text Reasoning & RAG Retrieval Flow

Unstructured citizen queries undergo regex filtering, LLM intent parsing, clarification check, vector retrieval, and verbatim facts assembling:

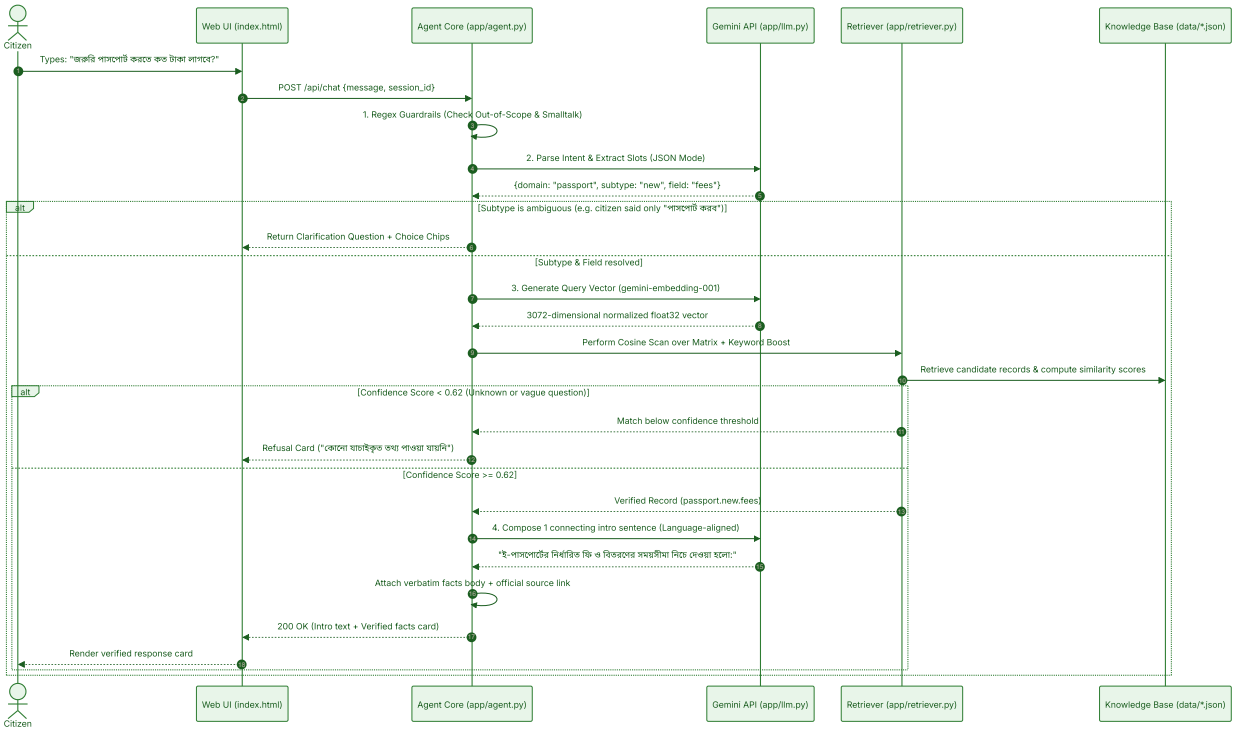


Figure 3: Cognitive RAG flow with strict hallucination barriers and confidence gating.

4. Mathematical Scoring & Retrieval Mechanism

Because the curated knowledge base consists of ~70 focused entries across 14 sub-types and 5 fields, an approximate-nearest-neighbor index (e.g. Chroma, FAISS, Pinecone) would introduce unnecessary cold-start latency and external operational dependencies.

Seba Sohayok implements an in-memory **NumPy vectorized cosine similarity scan** combined with a lexical keyword boost:

```
Score(q, d) = cos(e_q, e_d) + β × KeywordBoost(q, d)

where cos(e_q, e_d) = (e_q · e_d) / (||e_q|| × ||e_d||)
```

Where:

- `e_q`: 3072-dimensional embedding vector of the citizen query (via `gemini-embedding-001`).
- `e_d`: Pre-computed normalized embedding vector stored in `data/embeddings.npz`.
- `KeywordBoost(q, d)`: Exact substring matching weight covering multilingual keywords (Bangla, English, and phonetic Banglish).
- `β`: Heuristic boost factor ensuring high-frequency administrative terms reliably surface exact domain fields.

5. Component Breakdown & Module Matrix

Module	Component	Responsibility & Invariant Guarantees
`app/main.py`	FastAPI Application	HTTP routing, CORS negotiation, static file mounting, session dispatch, and OpenAPI schema exposure.
`app/agent.py`	Goal-Based Agent Core	State tracking, regex guardrails, clarification logic, answer card assembly, and multilingual translation stitching.
`app/retriever.py`	Vector Retrieval Engine	Direct NumPy matrix cosine scanning, keyword boosting, threshold filtering, and offline embedding index generation.
`app/llm.py`	Gemini Client Gateway	Official `google-genai` wrapper, structured JSON output enforcement, transient error retries, and rate handling.
`app/kb.py`	Taxonomy & Data Contract	Validates 14 sub-types and 5 slot fields; parses, normalizes, and loads JSON datasets into runtime memory.
`app/admin.py`	Human-in-the-Loop Admin	Drafts new entries from raw gazette text, manages `pending.json`, and updates KB on official human sign-off.
`app/limits.py`	Rate Limiting	In-memory per-IP token bucket preventing API quota exhaustion (20 req/min, 200 req/day).

6. Academic Formulation (Russell & Norvig PEAS Specification)

PEAS Dimension	System Specification
Performance Measure (P)	Zero hallucination rate on numerical/procedural facts; precision ≥ 95% at confidence threshold 0.62; turn latency < 800ms (free-text) and < 10ms (action chips).
Environment (E)	Bilingual public web client; asynchronous HTTP REST server; curated government administrative knowledge base.
Actuators (A)	HTTP JSON responses, structured UI cards, dynamic clarification suggestion chips, and verified external source citations.
Sensors (S)	Client HTTP requests (natural language strings, action identifiers, and session tokens).
Environment Properties	Partially Observable, Single-Agent, Sequential, Static (within turn), Discrete, and Predominantly Deterministic.

7. Knowledge Base Data Contract


```
{
  "id": "passport.new.fees",
  "domain": "passport",
  "subtype": "new",
  "field": "fees",
  "title_bn": "ই-পাসপোর্ট ফি ও বিতরণ সময়",
  "title_en": "e-Passport Fees & Delivery Timeframe",
  "content_bn": "৪৮ পৃষ্ঠা ৫ বছর মেয়াদ: নিয়মিত ৪,০২৫ টাকা, জরুরি ৬,৩২৫ টাকা...",
  "content_en": "48 pages 5 years: Regular 4,025 BDT, Urgent 6,325 BDT...",
  "source_url": "https://www.epassport.gov.bd/instructions/passport-fees",
  "verified_on": "2026-09-18",
  "keywords": ["ফি", "টাকা", "খরচ", "fee", "cost", "taka", "khoroch", "urgent"]
}
```